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The Concepts TS improves upon C++

Abstract

There is concern that the Concepts TS does not improve C++, and that it might not be necessary. This paper addresses some concerns, and considers why having and using (but not necessarily writing) concepts is beneficial to the average C++ programmer.

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Introduction

This paper addresses some of the concerns that some programmers have about whether the Concepts TS contributes to the C++ programming language.

Definitions

- **Modern C++** refers to any C++ Standard published since 2011.

- **Concepts TS** is shorthand for *ISO/IEC TS 19217:2015, Programming Languages – C++ Extensions for Concepts*.
- **Concepts** is shorthand for the Concepts TS
- **concept** refers to an individual `concept` written in code
- **Ranges TS** is shorthand for *N4685, Working Draft, C++ Extensions for Ranges*.
- **Ranges** is shorthand for the Ranges TS.
- **Novice C++ programmer** (or some variant) refers to a C++ programmer with less than three years experience with modern C++.
- **Average C++ programmer** (or some variant) refers to a C++ programmer with three or more years experience with modern C++.
- **Expert C++ programmer** (or some variant) refers to a C++ programmer that satisfies at least one of the following criteria:
 - i. attends WG21-related meetings (e.g. this meeting)
 - ii. attends WG21-related calls (e.g. SG14 telecons)
 - iii. contributes to Standard proposal papers
 - iv. contributes to a modern C++ compiler (e.g. GCC)
 - v. contributes to a modern C++ Standard Library (e.g. libcpp)
 - vi. contributes to a modern C++ library similar to the Standard Library (e.g. Boost, range-v3, etc.)
 - vii. contributes to a reference implementation for a Standard proposal or Technical Specification (e.g. cmcstl2)
 - viii. has ten or more years experience with C++, particularly with templates, including template metaprogramming (includes using Boost.MPL)
- **Clang** or **LLVM** refer to the C++ implementation of LLVM 5.
 - **libcpp** refers to the LLVM implementation of the C++ Standard Library, provided by the flag `-stdlib=libcpp` to Clang.
- **GCC** refers to the C++ implementation of GCC 7.
 - **libstdc++** refers to the GCC implementation of the C++ Standard Library.
- **MSVC** refers to the C++ implementation of Microsoft Visual C++ 2017.

All compiler diagnostics are provided through Matt Godbolt's *Compiler Explorer*, and are also present in Appendix A for offline viewing. Following the links to Compiler Explorer is advised.

Error messages with Concepts and Ranges

One particular concern is that Concepts do not improve compiler diagnostics. To illustrate, we'll consider the following two programs:

- Listing 1. [Without concepts](#)

```
#include <algorithm>
#include <iterator>
#include <list>

int main()
{
    auto l = std::list{1, 2, 3, 4, 5};
    std::sort(std::begin(l), std::end(l));
}
```

- Listing 2. [With concepts](#)

```
#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <list>

namespace ranges = std::experimental::ranges;

int main()
{
    auto l = std::list{1, 2, 3, 4, 5};
    ranges::sort(ranges::begin(l), ranges::end(l));
}
```

Both diagnostics – on their own and at a first glance – appear to be expert-friendly. Both are incredibly verbose, which can be intimidating, and the `ranges::sort` is sufficiently more verbose. This does not mean that it is sufficiently more expert-friendly than `std::sort`: instead, it opens opportunities for tools to provide more *informative* diagnostics. At present, this might be using a third-party tool that only reveals certain text from the diagnostic (e.g. showing only the parts that match the regular expression “concept bool\W+\w+\W*=”); in future, it might be possible for a compiler to implement a feature that cleans up the diagnostics further.

It's not possible to do this solely with template diagnostics because the template does not contain the necessary information. The information that is spewed from `std::sort` rambles on about an `operator-` that the user has not implemented. The only piece of information that is actually informative is `_RandomAccessIterator`, but this isn't easily findable when using libc++, and there is no mention of the word ‘random’ in the diagnostic from MSVC 2017 at all. Since all three major compilers discuss `operator-`, and only two of them mention ‘random’ (almost in passing), the following information can be inferred from the diagnostic:

1. It has something to do with `std::sort` (given that libstdc++ talks about `std::__sort`, libc++ talks about `std::__1::__insertion_sort`, and MSVC talks about both `std::sort` and `std::_Sort_unchecked`).
2. It has something to do with `operator-`

In contrast, `ranges::sort` spews diagnostics regarding the interface for `sort`. There is rambling, but it explicitly mentions a few key words:

1. `std::experimental::ranges::v1::sort`
2. `concept bool RandomAccessIterator`
3. `concept bool DerivedFrom`
4. `concept bool SizedSentinel`
5. `concept bool StrictTotallyOrdered`
6. `concept bool __totally_ordered`
7. `concept bool RandomAccessIncrementable`
8. `concept bool RandomAccessIterator`
9. `RandomAccessRange` and `Sortable`

Even most of this is noise, but we get a lot more information. Firstly, there's no mention of `operator-`, which is an *implementation detail* of the sort algorithm. This means that we are left to consider how we interact with our algorithm. The diagnostic provided by the concept also mentions the term `RandomAccess` four times: this means that we're likely violating some contract that asks for a random access data structure: this is a topic that is covered in the first year of computer science at university, or when `std::list` is introduced in a well-written resource (there's no need to discuss random access and linear access until a non-`vector` container appears).

As previously stated, the diagnostics can be improved upon here: what we currently have is a prototype.

The quality of diagnostics is a strong motivator for using concepts, but it's not the only reason to use them. There's more on that below. If the use of concepts served solely as a vehicle to improve diagnostics, the compiler might not be the tool to grok the diagnostics: we use other tools, such as IDEs, lint tools, and static analysers. Perhaps one of these can pick up when a compiler struggles to provide ample documentation. Recall that the Standard does not specify *how* diagnostics are to be output – only that they are – and users who prefer not to use tools can request better diagnostics from their compiler vendor.

Let's now consider sorting on a type that can't be compared.

- Listing 3. [Without concepts](#)

```
#include <algorithm>
#include <iterator>
#include <vector>

class Foo {};

int main()
{
    auto l = std::vector<Foo>{};
    std::sort(std::begin(l), std::end(l));
}
```

- Listing 4. [With concepts](#)

```
#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <vector>

namespace ranges = std::experimental::ranges;

class Non_sortable {};

int main()
{
    auto l = std::vector<Non_sortable>{};
    ranges::sort(ranges::begin(l), ranges::end(l));
}
```

Toward the top of both libstdc++ and libc++, we can see that an expression of the form `a < b` is invalid. MSVC also mentions `std::less<void>`, so we can safely conclude that all three major implementations of the Standard Library communicate we're falling flat on our faces with a less-than comparison. Beyond that, there isn't terribly much useful information, and it isn't clear from the way that it is presented that a lacking `bool operator<(const Foo&, const Foo&)` is the culprit:

1. GCC talks a lot about `{ return *__it1 < *__it2; }`. This snippet of code exposes the implementation of `std::sort` to users. Users shouldn't be concerned with a correct implementation of `std::sort`; they should be focusing on the contract that they've ignored.
2. LLVM is exempt, its diagnostic is rather clear.
3. MSVC's diagnostics are less clear, but with some squinting, we can see that they are related to `Foo`.

The Ranges TS provides vastly different output. The most relevant part of the diagnostic is

```

concept bool Sortable =
concept bool IndirectStrictWeakOrder =
concept bool StrictWeakOrder =
concept bool Relation =
concept bool StrictWeakOrder =
concept bool Relation =
concept bool StrictWeakOrder =
concept bool Relation =
concept bool StrictWeakOrder =
concept bool Relation =
concept bool StrictWeakOrder =
concept bool Relation =

```

To understand what is happening here, we need to understand what it means to be `Sortable`, what a `StrictWeakOrder` is, and what a `Relation` is. We can look to mathematics for the definition of both a relation[3][4] and a strict weak order[5][6]. (We discuss `Sortable` in detail later).

`StrictWeakOrder` is also known as the [Compare concept](#) in the C++ Standard Library.

While it requires programmers to know more terms, these terms remain consistent with the C++ Standard, as opposed to the diagnostics returned from something such as `std::sort`, which may vary from implementation to implementation.

Generic programming with Concepts

It has been argued that real-world concepts, such as `Sortable`, are difficult to understand. This is correct. I do not expect the authors of the Palo Alto TR, nor the authors of the Ranges TS to disagree either. The mathematics involved is complex and took years to correctly derive (the Palo Alto TR was published in 2012; the Ranges TS in 2017). The algebra required to employ Standard algorithms is present regardless of them using concepts – wildly incorrect code will still violate the contract, and well-specified concepts can help to *enforce* said contract.

What the Ranges TS does is provide a *formal* way for compilers (and other tools) to check if a type can be sorted, and *clearly* communicate the requirements of an interface to users. Someone needs to write `Sortable` – in fact, it took at least two people to define this, possibly more – which means that it's not a trivial requirement to properly pin down. Leaving this in the hands of the average programmer is not advisable.

Concepts should not be *written* by average programmers: they only need to *use* concepts to reap the benefits of such complex code. A textbook or reference might summarise `Sortable` as

An iterator is `Sortable` if, and only if, it is mutable and its `value_type` is both `Movable` and has a strict weak ordering imposed on it.

The terms “mutable”, `Movable`, and “strict weak ordering” are all fairly simple to communicate, and would ideally be introduced well before the definition of the term `Sortable` in a well-designed course or textbook.

One might wonder if concepts are simple to compose for experts. Again, they are not, and should not be trivially composable. A descriptive concept isn’t written for trivial things such as `Addable`, `Subtractable`, etc.: `Sortable` aims to mathematically capture the definition of a sortable type. Spending five years on modelling the requirements for the Standard Library implicitly suggests that concepts *should not* be an easy pass.

This demonstrates that even experts shouldn’t be writing concepts on their own. They can use them, knowing that once they’ve been thoroughly vetted, the code is guaranteed to block things that shouldn’t be used as parameters.

Generic programming in C++17

Modern C++ offers various tools for expressive generic programming, and some people wonder if these make concepts redundant. We’ll consider them in turn.

1. `std::enable_if`
2. Tag dispatching
3. `constexpr - if` statements
4. `constexpr bool`

`std::enable_if`

`enable_if` is a hacky way to implement concepts at best. `enable_if<Condition, T>` will have a member alias called `type` if, and only if, `Condition` evaluates to `true`. This provides a way to leverage SFINAE to conditionally disable template specialisations and overloads. However, diagnostics will range from [overload resolution failures](#) to [incomplete type errors](#) none of which provide information of *why* there was a failure.

Concepts, on the other hand, offer a robust mechanism for detecting which requirement isn’t satisfied. This is likely due to the fact that it is a first-class, core language feature. Another bonus is that it explicitly communicates that a constraint hasn’t been satisfied – something that is not made clear by our `enable_if` hack!

This can be demonstrated by taking a look at the diagnostics from the following examples.

Not `EqualityComparable`

- Listing 5. [Without concepts](#)

```
#include <regex>
#include <type_traits>

template <typename T>
constexpr bool Destructible = std::is_destructible_v<T>;

template <typename T>
constexpr bool DefaultConstructible = Destructible<T> && std::is_default_constructible_v<T>;

template <typename T>
constexpr bool Movable = Destructible<T> && std::is_move_constructible_v<T> && std::is_move_assignable_v<T>;

template <typename T>
constexpr bool Copyable = Destructible<T> && std::is_copy_constructible_v<T> && std::is_copy_assignable_v<T>;

template <typename T>
constexpr bool Semiregular = Copyable<T> && DefaultConstructible<T>;

template <typename T>
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>(), bool> &&
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>(), bool>;

template <typename T>
constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;

int main()
{
    std::enable_if_t<Regular<std::regex>, std::regex> foo{};
}
```

- Listing 6. [With concepts](#)

```
#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <regex>

namespace ranges = std::experimental::ranges;
using ranges::Regular;

int main()
{
    Regular foo = std::regex{};
}
```

While it's possible to work out that `std::regex` isn't `EqualityComparable` when using GCC (without concepts), it isn't nearly as clear as the concept-equivalent GCC implementation. There are considerable more diagnostics to decipher, and GCC sells the idea that the problem lies with the definition of `EqualityComparable`, instead of directing the programmer to understand that `std::regex` fails to meet the requirements of `Regular`, because it fails to meet the requirements outlined by `EqualityComparable`.

MSVC identifies that `std::regex` doesn't have an `operator==`, nor an `operator!=`, but it also spits out a lot of spurious and irrelevant diagnostics.

Clang offers the best non-concept-diagnostics. It also sells the idea that there is a problem with `EqualityComparable`, but words it in such a way that with experience, one can be able to discern what is happening. The biggest issue is that it reports the error in reverse – something that Concepts fixes.

Not `DefaultConstructible`

- Listing 7. [Without concepts](#)

```
#include <regex>
#include <type_traits>

class Foo {
public:
    Foo() = delete;
};

constexpr bool operator==(const Foo, const Foo) noexcept;
constexpr bool operator!=(const Foo, const Foo) noexcept;

template <typename T>
constexpr bool Destructible = std::is_destructible_v<T>;

template <typename T>
constexpr bool DefaultConstructible = Destructible<T> && std::is_default_constructible_v<T>;

template <typename T>
constexpr bool Movable = Destructible<T> && std::is_move_constructible_v<T> && std::is_copy_constructible_v<T>;

template <typename T>
constexpr bool Copyable = Destructible<T> && std::is_copy_constructible_v<T> && std::is_move_constructible_v<T>;

template <typename T>
constexpr bool Semiregular = Copyable<T> && DefaultConstructible<T>;

template <typename T>
```

```
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>() == std::declval<T>()), bool>;

template <typename T>
constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;

int main()
{
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
}
```

- Listing 8. [With concepts](#)

```
#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <regex>

namespace ranges = std::experimental::ranges;
using ranges::Regular;

class Foo {
public:
    Foo() = delete;
};

constexpr bool operator==(const Foo, const Foo) noexcept;
constexpr bool operator!=(const Foo, const Foo) noexcept;

int main()
{
    Regular foo = Foo{};
}
```

This is a fairly open-shut case. None of the Concepts-absent compilers report that `Foo` is not a `Regular` type because it is not `DefaultConstructible`. The Concepts-enabled GCC implementation does.

◀ **Not `DefaultConstructible` and not `Copyable`** ▶

- Listing 9. [Without concepts](#)

```
#include <regex>
#include <type_traits>

class Foo {
public:
```

```

    Foo() = delete;
    Foo(const Foo&) = delete;
};

constexpr bool operator==(const Foo&, const Foo&) noexcept;
constexpr bool operator!=(const Foo&, const Foo&) noexcept;

template <typename T>
constexpr bool Destructible = std::is_destructible_v<T>;

template <typename T>
constexpr bool DefaultConstructible = Destructible<T> && std::is_default_constructible_v<T>;

template <typename T>
constexpr bool Movable = Destructible<T> && std::is_move_constructible_v<T> && std::is_move_assignable_v<T>;

template <typename T>
constexpr bool Copyable = Destructible<T> && std::is_copy_constructible_v<T> && std::is_copy_assignable_v<T>;

template <typename T>
constexpr bool Semiregular = Copyable<T> && DefaultConstructible<T>;

template <typename T>
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()), decltype(std::declval<T>())>() == std::is_same_v<decltype(std::declval<T>()), decltype(std::declval<T>())>(), bool>;

template <typename T>
constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;

int main()
{
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
}

```

- Listing 10. [With concepts](#)

```

#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <regex>

namespace ranges = std::experimental::ranges;
using ranges::Regular;

class Foo {
public:
    Foo() = delete;
    Foo(const Foo&) = delete;
};

```

```
constexpr bool operator==(const Foo&, const Foo&) noexcept;
constexpr bool operator!=(const Foo&, const Foo&) noexcept;

int main()
{
    Regular foo = Foo{};
}
```

Again, non-Concepts compilers don't identify why the code is ill-formed. There are a lot of diagnostics from the Concepts-enabled GCC – perhaps this can be worked on, but at least it identifies the problem.

Tag dispatching

- Listing 11. [Without concepts](#)

```
#include <iterator>

template <typename I>
void advance_helper(I i, typename std::iterator_traits<I>::difference_type n, std::ra
{
    i += n;
}

template <typename I>
void advance_helper(I i, typename std::iterator_traits<I>::difference_type n, std::bi
{
    for (; n > 0; --n)
        ++i;
    for (; n < 0; ++n)
        --i;
}

template <typename I>
void advance_helper(I i, typename std::iterator_traits<I>::difference_type n, std::in
{
    for (; n > 0; --n)
        ++i;
}

template <typename I>
void advance(I i, typename std::iterator_traits<I>::difference_type n)
{
    advance_helper(i, n, typename std::iterator_traits<I>::iterator_category{});
}
```

- Listing 12. [With concepts](#)

```
#include <experimental/ranges/iterator>

namespace ranges = std::experimental::ranges;

template <ranges::RandomAccessIterator I>
void advance_helper(I i, ranges::difference_type_t<I> n)
{
    i += n;
}

template <ranges::BidirectionalIterator I>
void advance_helper(I i, ranges::difference_type_t<I> n)
{
    for (; n > 0; --n)
        ++i;
    for (; n < 0; ++n)
        --i;
}

template <ranges::InputIterator I>
void advance_helper(I i, ranges::difference_type_t<I> n)
{
    for (; n > 0; --n)
        ++i;
}

template <typename I>
void advance(I i, ranges::difference_type_t<I> n)
{
    advance_helper(i, n);
}
```

Listing 11 is a slightly modified version of this example implementation[7] of `std::advance` to illustrate type traits. This isn't exactly complex code, but it is clunky, and it's possible to trick the compiler into thinking that we have a tag that we don't really have. It is also very awkward to call this function: we need to define an object that doesn't really serve any purpose other than to tell the compiler "call this function".

Listing 12 eliminates all of these problems. It clearly states what the requirements are in the template header and makes it considerably harder to trick the compiler. Furthermore, we don't need to have a tag to say "call me!". This makes *using* constrained functions simpler than their tag-counterparts.

constexpr - if statements

- Listing 13. [Replacing concepts with `constexpr if`](#)

```
#include <experimental/ranges/iterator>

namespace ranges = std::experimental::ranges;

template <typename I>
void advance(I i, ranges::difference_type_t<I> n)
{
    if constexpr (ranges::RandomAccessIterator<I>) {
        i += n;
    }
    else if constexpr (ranges::BidirectionalIterator<I>) {
        for (; n > 0; --n)
            ++i;
        for (; n < 0; ++n)
            --i;
    }
    else {
        for (; n > 0; --n)
            ++i;
    }
}
```

- Listing 14. [Appropriate use of `constexpr if`](#)

```
#include <experimental/ranges/concepts>
#include <fstream>
#include <string>
#include <deque>

namespace ranges = std::experimental::ranges;

template <template <typename...> typename C, typename T>
C<T> from_file(const std::string& path)
{
    if (auto in = std::ifstream{path}) {
        const ranges::SignedIntegral size = [&in]{
            ranges::SignedIntegral i = 0;
            in >> i;
            return i;
        }();

        auto c = [size]{
            auto c = C<T>{};
            if constexpr (std::is_same_v<C<T>, std::vector<T>>)
                c.reserve(size);
        };
    }
}
```

```

        return c;
    }();
    // ...

    return c;
}

int main()
{
    from_file<std::vector, int>("foo");
}

```

`constexpr if` statements are a great way to eliminate SFINAE, as are Concepts. One might argue that the above program could also be written as it is in Listing 13. This is terrible design. We have combined three different functions, which articulate three different forms of logic, into a single function that operates on `if`-statements.

`constexpr if`-statements are a powerful tool, that when used correctly, are a great way for avoiding nasty and obscure SFINAE. This is great in small doses, but it isn't viable for something where lots of specialised logic is present, or the logic for a specialisation is independent to another specialisation.

An example of what constitutes as a 'valid' use of `constexpr if` might look like Listing 14.

The point of this contrived example is to show that `if constexpr` is used to complement the function: not to completely switch behaviour on a different type, as is demonstrated in the `advance` port.

This is an example of separating interfaces from implementations. Listing 13 confuses the boundary between the interface and the implementation that is very clearly provided in Listing 12. Listing 14 does no such thing – enables one features that are very specific to `std::vector`, but it in no way alters the bulk of the algorithm if we choose to use `std::deque` instead.

constexpr bool objects

Some might argue that the constrained variable declaration syntax can already be achieved by templated `constexpr bool` objects. For an example, simply visit any of the links to Compiler Explorer for a non-Concepts-enabled example.

When using them with `enable_if`, they choke out the ability to use automatic type deduction: perhaps we should write code like so:

```

auto v = std::vector<int>{}; static_assert(Regular<decltype(v)>);

```

Quite frankly, this is far more intrusive than needing to learn about Concepts' constrained variable syntax, which enables this check, by default; additionally, I wager that it would take a very dedicated programmer to provide concept checks in such a manner. Then, the code will need to pass a review, and reviewers mightn't like such code, or understand why it's necessary. Arguing that the regularity is a part of the type deduction is far simpler, and far more readable.

This also doesn't take contexts where a `static_assert` can't be used directly after the declaration into consideration. For example, a range-for statement:

```
for (auto i : v) {  
    static_assert(Regular<decltype(i)>);  
    // ...  
}
```

And we haven't looked at reference types. To check that the underlying type of a reference is regular, we ought to strip the reference part, because `Regular<T&>` doesn't mean the same thing as `Regular<T>`. To do that, we need to use `std::decay_t`. Or is it `std::remove_reference_t`? What about when `const` is involved? That's `std::remove_const_t`. Should I just combine all of them together and hope for the best? Crap, the order matters too.

`std::remove_reference_t<std::remove_const_t<T>>` isn't the same as `std::remove_const_t<std::remove_reference_t<T>>`. Where does `std::decay` fit in again? Is this snippet right? Is it *understandable*?

```
for (const auto& i : v) {  
    static_assert(Regular<std::remove_const_t<std::remove_reference_t<decltype(i)>>>>);  
    // ...  
}
```

Concepts avoid this issue by going directly to the underlying type, which is checked before the reference is ever applied. We can use

```
for (const Regular& i : v) {  
    // ...  
}
```

and *not* worry about references or `const` types.

Modern C++ programmers shouldn't need to worry about template metaprogramming on a daily basis. Novices should worry about template metaprogramming even less than average C++ programmers. The problem is that we thrust template metaprogramming on to non-experts and non-

library developers in contexts that probably do not require such code. Concepts, especially when paired with Ranges, offer a way to curb high amounts of template metaprogramming. Eric Niebler's [Range-v3](#) library offers no such mechanism; it does not have language support.

Conclusion

There are arguments that the Concepts TS don't offer anything to simplify generic programming in C++, but most overlook the fact that the tools we currently have are Hacky-at-Best™. Arguments made against diagnostics forget that there is only one reference implementation for Concepts, and that the “complex” diagnostics resulting from said implementation can always be improved upon.

Concepts do simplify generic programming when used correctly. As someone who works with templates on a daily basis, tools such as `enable_if` are obscure and tools such as `constexpr if` are not always the correct answer. To add concepts to the mix is to add:

- a valuable tool that helps reduce the amount of expert-friendly code by banishing complex template metaprogramming to concept definitions,
- first-class support for constraint-based type deduction
- offering the opportunity to receive clearer diagnostics
- the choice between `constexpr if` and constrained functions when the situation calls for one or the other.

Concepts don't make any of the above features redundant; nor do any of the above features make Concepts dead-on-arrival. They instead offer programmers the opportunity to choose the correct tool for the correct problem. Concepts make C++ better.

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Appendix A

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Listing 1

Source

```
#include <algorithm>
#include <iterator>
```

```
#include <list>
```

```
int main()
{
    auto l = std::list{1, 2, 3, 4, 5};
    std::sort(std::begin(l), std::end(l));
}
```

GCC diagnostic

```
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:62
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h: In instantiation
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:4836:18:   require
8 : <source>:8:39:   required from here
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1969:22: error: no
      std::__lg(__last - __first) * 2,
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:389:5: note: c
      operator-(const reverse_iterator<_IteratorL>& __x,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:389:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:62
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1969:22: note:   '
      std::__lg(__last - __first) * 2,
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1191:5: note:
      operator-(const move_iterator<_IteratorL>& __x,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1191:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:62
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1969:22: note:   '
      std::__lg(__last - __first) * 2,
      ~~~~~^~~~~~
Compiler exited with result code 1
```

Clang diagnostic

```

In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4007:40: error: in
    difference_type __len = __last - __first;
                        ~~~~~ ^ ~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<_Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<_RandomAccessIterato
    ^
8 : <source>:8:8: note: in instantiation of function template specialization 'std::__
    std::sort(std::begin(l), std::end(l));
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:747:1: note: candid
operator-(const reverse_iterator<_Iter1>& __x, const reverse_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1166:1: note: candi
operator-(const move_iterator<_Iter1>& __x, const move_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1571:1: note: candi
operator-(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter2>& __y) _NOEXCEPT_D
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4098:33: error: in
    if (__i >= __j)
        ~~~ ^ ~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/utility:583:1: note: candida
operator>=(const pair<_T1,_T2>& __x, const pair<_T1,_T2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:730:1: note: candid
operator>=(const reverse_iterator<_Iter1>& __x, const reverse_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1149:1: note: candi
operator>=(const move_iterator<_Iter1>& __x, const move_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1522:1: note: candi
operator>=(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter2>& __y) _NOEXCEPT_
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1554:1: note: candi
operator>=(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter1>& __y) _NOEXCEPT_
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/tuple:1203:1: note: candidat
operator>=(const tuple<_Tp...>& __x, const tuple<_Up...>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2889:1: note: candida
operator>=(const unique_ptr<_T1, _D1>& __x, const unique_ptr<_T2, _D2>& __y) {return
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2976:1: note: candida
operator>=(const unique_ptr<_T1, _D1>& __x, nullptr_t)

```

```

^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2984:1: note: candidate
operator>=(nullptr_t, const unique_ptr<_T1, _D1>& __x)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4721:1: note: candidate
operator>=(const shared_ptr<_Tp>& __x, const shared_ptr<_Up>& __y) _NOEXCEPT
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4809:1: note: candidate
operator>=(const shared_ptr<_Tp>& __x, nullptr_t) _NOEXCEPT
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4817:1: note: candidate
operator>=(nullptr_t, const shared_ptr<_Tp>& __x) _NOEXCEPT
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/list:2396:1: note: candidate
operator>=(const list<_Tp, _Alloc>& __x, const list<_Tp, _Alloc>& __y)
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4122:17: error: in
    if (__i < __j)
        ~~~ ^ ~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/utility:567:1: note: candidate
operator< (const pair<_T1,_T2>& __x, const pair<_T1,_T2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:706:1: note: candidate
operator<(const reverse_iterator<_Iter1>& __x, const reverse_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1125:1: note: candidate
operator<(const move_iterator<_Iter1>& __x, const move_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1494:1: note: candidate
operator<(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter2>& __y) _NOEXCEPT_D
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/tuple:1187:1: note: candidate
operator<(const tuple<_Tp...>& __x, const tuple<_Up...>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2868:1: note: candidate
operator< (const unique_ptr<_T1, _D1>& __x, const unique_ptr<_T2, _D2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2926:1: note: candidate
operator<(const unique_ptr<_T1, _D1>& __x, nullptr_t)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2935:1: note: candidate
operator<(nullptr_t, const unique_ptr<_T1, _D1>& __x)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4696:1: note: candidate
operator<(const shared_ptr<_Tp>& __x, const shared_ptr<_Up>& __y) _NOEXCEPT
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4761:1: note: candidate
operator<(const shared_ptr<_Tp>& __x, nullptr_t) _NOEXCEPT

```

```

^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4769:1: note: candidate
operator<(nullptr_t, const shared_ptr<_Tp>& __x) _NOEXCEPT
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/list:2372:1: note: candidate
operator<(const list<_Tp, _Alloc>& __x, const list<_Tp, _Alloc>& __y)
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4156:65: error: in
    if (_VSTD::__insertion_sort_incomplete<_Compare>(__i+1, __last, __comp))
                                                ~~~^~

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:765:1: note: candid
operator+(typename reverse_iterator<_Iter>::difference_type __n, const reverse_iterat
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1184:1: note: candi
operator+(typename move_iterator<_Iter>::difference_type __n, const move_iterator<_It
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1597:1: note: candi
operator+(typename __wrap_iter<_Iter>::difference_type __n,
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4173:17: error: in
    if (__i - __first < __last - __i)
        ~~~ ^ ~~~~~~

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:747:1: note: candid
operator-(const reverse_iterator<_Iter1>& __x, const reverse_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1166:1: note: candi
operator-(const move_iterator<_Iter1>& __x, const move_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1571:1: note: candi
operator-(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter2>& __y) _NOEXCEPT_D
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3915:20: error: in
    switch (__last - __first)
        ~~~~~ ^ ~~~~~~

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4155:32: note: in
    bool __fs = _VSTD::__insertion_sort_incomplete<_Compare>(__first, __i, __
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<_Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<_RandomAccessIterato
    ^
8 : <source>:8:8: note: in instantiation of function template specialization 'std::__
    std::sort(std::begin(l), std::end(l));
    ^

```

```

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:747:1: note: candid
operator-(const reverse_iterator<_Iter1>& __x, const reverse_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1166:1: note: candi
operator-(const move_iterator<_Iter1>& __x, const move_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1571:1: note: candi
operator-(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter2>& __y) _NOEXCEPT_D
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3935:40: error: in
    _RandomAccessIterator __j = __first+2;
                        ~~~~~~^~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:765:1: note: candid
operator+(typename reverse_iterator<_Iter>::difference_type __n, const reverse_iterat
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1184:1: note: candi
operator+(typename move_iterator<_Iter>::difference_type __n, const move_iterator<_It
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1597:1: note: candi
operator+(typename __wrap_iter<_Iter>::difference_type __n,
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3936:39: error: in
    __sort3<_Compare>(__first, __first+1, __j, __comp);
                ~~~~~~^~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:765:1: note: candid
operator+(typename reverse_iterator<_Iter>::difference_type __n, const reverse_iterat
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1184:1: note: candi
operator+(typename move_iterator<_Iter>::difference_type __n, const move_iterator<_It
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1597:1: note: candi
operator+(typename __wrap_iter<_Iter>::difference_type __n,
^
8 errors generated.
Compiler exited with result code 1

```

MSVC diagnostic

```

example.cpp
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/xlocale(314): warning C
7 : <source>(7): error C2955: 'std::list': use of class template requires template ar
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/list(879): note: see de
8 : <source>(8): error C3536: 'l': cannot be used before it is initialized
8 : <source>(8): error C2672: 'std::begin': no matching overloaded function found
8 : <source>(8): error C2893: Failed to specialize function template 'unknown-type st

```

```

8 : <source>(8): note: With the following template arguments:
8 : <source>(8): note: '_Container=int'
8 : <source>(8): error C2784: 'const _Elem *std::begin(std::initializer_list<_Elem>)' no matching function found
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/initializer_list(61): note:
8 : <source>(8): error C2672: 'std::end': no matching overloaded function found
8 : <source>(8): error C2893: Failed to specialize function template 'unknown-type st
8 : <source>(8): note: With the following template arguments:
8 : <source>(8): note: '_Container=int'
8 : <source>(8): error C2784: 'const _Elem *std::end(std::initializer_list<_Elem>)' no matching function found
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/initializer_list(68): note:
8 : <source>(8): error C2672: 'std::sort': no matching overloaded function found
8 : <source>(8): error C2780: 'void std::sort(_RanIt,_RanIt)': expects 2 arguments -
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2921): note:
8 : <source>(8): error C2780: 'void std::sort(_RanIt,_RanIt,_Pr)': expects 3 arguments
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2913): note:
Microsoft (R) C/C++ Optimizing Compiler Version 19.10.25017 for x64
Copyright (C) Microsoft Corporation. All rights reserved.
Compiler exited with result code 2

```

Listing 2

Source

```

#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <list>

namespace ranges = std::experimental::ranges;

int main()
{
    auto l = std::list{1, 2, 3, 4, 5};
    ranges::sort(ranges::begin(l), ranges::end(l));
}

```

GCC diagnostic

```

<source>: In function 'int main()':
10 : <source>:10:48: error: no matching function for call to 'sort(std::_List_iterator, std::_List_iterator)'
    ranges::sort(ranges::begin(l), ranges::end(l));
                  ^
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:31:4: note:

```

```

I sort(I first, S sent, Comp comp = Comp{}, Proj proj = Proj{})
    ^~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:31:4: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/iterator.hpp:2
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:652:15:
    concept bool RandomAccessIterator =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:76:15: note
    concept bool DerivedFrom =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:76:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:76:15: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/iterator.hpp:2
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:545:15:
    concept bool SizedSentinel =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:545:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:545:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:545:15:
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:140:15:
    concept bool StrictTotallyOrdered =
        ^~~~~~

```

```

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
    concept bool __totally_ordered =
        ^~~~~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:125:15:
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/iterator.hpp:2
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
    concept bool RandomAccessIncrementable =
        ^~~~~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/increment.hpp:138:16
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterato
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/iterator.hpp:2
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:652:15:
    concept bool RandomAccessIterator =
        ^~~~~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:652:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/iterator/concepts.hpp:652:15:
10 : <source>:10:48: note: ill-formed constraint
    ranges::sort(ranges::begin(l), ranges::end(l));
        ^
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:46:23: not
    safe_iterator_t<Rng> sort(Rng&& rng, Comp comp = Comp{}, Proj proj = Proj{})

```

^~~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:46:23: not
Compiler exited with result code 1

Listing 3

Source

```
#include <algorithm>
#include <iterator>
#include <vector>

class Foo {};

int main()
{
    auto l = std::vector<Foo>{};
    std::sort(std::begin(l), std::end(l));
}
```

GCC diagnostic

```
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h: In instanti
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:81:17:   required
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1921:34:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1953:38:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1968:25:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:4836:18:   require
10 : <source>:10:39:   required from here
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:43:23: error
      { return *__it1 < *__it2; }
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:888:5: note: c
      operator<(const __normal_iterator<_IteratorL, _Container>& __lhs,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:888:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from <source>:1:
```

```

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:43:23: note:
    { return *__it1 < *__it2; }
        ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:895:5: note: c
    operator<(const __normal_iterator<_Iterator, _Container>& __lhs,
        ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:895:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:43:23: note:
    { return *__it1 < *__it2; }
        ~~~~~^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h: In instanti
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1828:20:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1855:36:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1885:25:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1971:31:   require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:4836:18:   require
10 : <source>:10:39:   required from here
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:90:22: error
    { return __val < *__it; }
        ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:888:5: note: c
    operator<(const __normal_iterator<_IteratorL, _Container>& __lhs,
        ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:888:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:90:22: note:
    { return __val < *__it; }
        ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:895:5: note: c
    operator<(const __normal_iterator<_Iterator, _Container>& __lhs,
        ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:895:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:

```

```

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:90:22: note:
    { return __val < *__it; }
        ~~~~~^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h: In instanti
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_heap.h:133:48: required
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_heap.h:237:23: required
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_heap.h:342:22: required
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1672:23: require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1933:25: require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1948:27: require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:1968:25: require
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_algo.h:4836:18: require
10 : <source>:10:39: required from here
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:65:22: error
    { return *__it < __val; }
        ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:888:5: note: c
    operator<(const __normal_iterator<_IteratorL, _Container>& __lhs,
        ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:888:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:65:22: note:
    { return *__it < __val; }
        ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:895:5: note: c
    operator<(const __normal_iterator<_Iterator, _Container>& __lhs,
        ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:895:5: note:
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/predefined_ops.h:65:22: note:
    { return *__it < __val; }
        ~~~~~^~~~~~

Compiler exited with result code 1

```

Clang diagnostic

```

In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:719:71: error: inv
    bool operator()(const _T1& __x, const _T1& __y) const {return __x < __y;}
                                   ~~~ ^ ~~~

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4014:17: note: in
    if (__comp(*--__last, *__first))
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<_Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<_RandomAccessIterato
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4225:12: note: in
    _VSTD::sort(__first.base(), __last.base());
        ^
10 : <source>:10:8: note: in instantiation of function template specialization 'std::
    std::sort(std::begin(l), std::end(l));
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/utility:567:1: note: candida
operator< (const pair<_T1,_T2>& __x, const pair<_T1,_T2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:706:1: note: candid
operator<(const reverse_iterator<_Iter1>& __x, const reverse_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1125:1: note: candi
operator<(const move_iterator<_Iter1>& __x, const move_iterator<_Iter2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/iterator:1494:1: note: candi
operator<(const __wrap_iter<_Iter1>& __x, const __wrap_iter<_Iter2>& __y) _NOEXCEPT_D
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/tuple:1187:1: note: candidat
operator<(const tuple<_Tp...>& __x, const tuple<_Up...>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2868:1: note: candida
operator< (const unique_ptr<_T1, _D1>& __x, const unique_ptr<_T2, _D2>& __y)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2926:1: note: candida
operator<(const unique_ptr<_T1, _D1>& __x, nullptr_t)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:2935:1: note: candida
operator<(nullptr_t, const unique_ptr<_T1, _D1>& __x)
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4696:1: note: candida
operator<(const shared_ptr<_Tp>& __x, const shared_ptr<_Up>& __y) _NOEXCEPT
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4761:1: note: candida
operator<(const shared_ptr<_Tp>& __x, nullptr_t) _NOEXCEPT
^

```

```

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/memory:4769:1: note: candidate
operator<(nullptr_t, const shared_ptr<_Tp>& __x) _NOEXCEPT
^
In file included from <source>:1:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3801:20: error: no
    unsigned __r = __sort3<_Compare>(__x1, __x2, __x3, __c);
                        ^~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4021:20: note: in
    _VSTD::__sort4<_Compare>(__first, __first+1, __first+2, --__last, __comp)
                        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<_Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<_RandomAccessIterato
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4225:12: note: in
    _VSTD::sort(__first.base(), __last.base());
    ^
10 : <source>:10:8: note: in instantiation of function template specialization 'std::
    std::sort(std::begin(l), std::end(l));
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3761:1: note: cand
__sort3(_ForwardIterator __x, _ForwardIterator __y, _ForwardIterator __z, _Compare __
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3827:20: error: no
    unsigned __r = __sort4<_Compare>(__x1, __x2, __x3, __x4, __c);
                        ^~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4024:20: note: in
    _VSTD::__sort5<_Compare>(__first, __first+1, __first+2, __first+3, --__la
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<_Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<_RandomAccessIterato
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4225:12: note: in
    _VSTD::sort(__first.base(), __last.base());
    ^
10 : <source>:10:8: note: in instantiation of function template specialization 'std::
    std::sort(std::begin(l), std::end(l));
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3798:1: note: cand
__sort4(_ForwardIterator __x1, _ForwardIterator __x2, _ForwardIterator __x3,
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3892:5: error: no
    __sort3<_Compare>(__first, __first+1, __j, __comp);
    ^~~~~~

```

```

/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4029:20: note: in
    _VSTD::__insertion_sort_3<Compare>(__first, __last, __comp);
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<RandomAccessIterato
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4225:12: note: in
    _VSTD::sort(__first.base(), __last.base());
        ^
10 : <source>:10:8: note: in instantiation of function template specialization 'std::
    std::sort(std::begin(l), std::end(l));
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3761:1: note: cand
__sort3(_ForwardIterator __x, _ForwardIterator __y, _ForwardIterator __z, _Compare __
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3925:9: error: no
    _VSTD::__sort3<Compare>(__first, __first+1, --__last, __comp);
    ^~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/__config:441:15: note: expan
#define _VSTD std::_LIBCPP_NAMESPACE
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4155:32: note: in
    bool __fs = _VSTD::__insertion_sort_incomplete<Compare>(__first, __i, __
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4200:5: note: in i
    __sort<Comp_ref>(__first, __last, __comp);
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4209:12: note: in
    _VSTD::sort(__first, __last, __less<typename iterator_traits<RandomAccessIterato
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:4225:12: note: in
    _VSTD::sort(__first.base(), __last.base());
        ^
10 : <source>:10:8: note: in instantiation of function template specialization 'std::
    std::sort(std::begin(l), std::end(l));
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3761:1: note: cand
__sort3(_ForwardIterator __x, _ForwardIterator __y, _ForwardIterator __z, _Compare __
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3928:9: error: no
    _VSTD::__sort4<Compare>(__first, __first+1, __first+2, --__last, __comp);
    ^~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/__config:441:15: note: expan
#define _VSTD std::_LIBCPP_NAMESPACE
        ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3798:1: note: cand

```

```

__sort4(_ForwardIterator __x1, _ForwardIterator __x2, _ForwardIterator __x3,
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3931:9: error: no
    _VSTD::__sort5<_Compare>(__first, __first+1, __first+2, __first+3, --__last,
    ^~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/__config:441:15: note: expanded from here
#define _VSTD std::_LIBCPP_NAMESPACE
    ^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3824:1: note: candidate function not viable: no known conversion from 'const _ForwardIterator' to 'const _ForwardIterator' for 1st argument
__sort5(_ForwardIterator __x1, _ForwardIterator __x2, _ForwardIterator __x3,
^
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3936:5: error: no
    __sort3<_Compare>(__first, __first+1, __j, __comp);
    ^~~~~~
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/algorithm:3761:1: note: candidate function not viable: no known conversion from 'const _ForwardIterator' to 'const _ForwardIterator' for 1st argument
__sort3(_ForwardIterator __x, _ForwardIterator __y, _ForwardIterator __z, _Compare __
^
8 errors generated.
Compiler exited with result code 1

```

MSVC diagnostic

```

example.cpp
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/xlocale(314): warning C
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2823): error
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2880): note:
    with
    [
        _RanIt=Foo *,
        _Pr=std::less<void>
    ]
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2908): note:
    with
    [
        _RanIt=Foo *,
        _Pr=std::less<void>,
        _Diff=__int64
    ]
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2916): note:
    with
    [
        _Pr=std::less<void>,
        _RanIt=Foo *
    ]
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2923): note:
    with
    [

```

[illegible]

```

    with
    [
        _RanIt=Foo *,
        _Pr=std::less<void>,
        _BidIt=Foo *
    ]
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2766): error
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2766): note:
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2766): note:
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2766): note:
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2774): error
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2774): error
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2774): note:
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2774): note:
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/algorithm(2774): note:
Microsoft (R) C/C++ Optimizing Compiler Version 19.10.25017 for x64
Copyright (C) Microsoft Corporation. All rights reserved.
Compiler exited with result code 2

```

Listing 4

Source

```

#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <vector>

namespace ranges = std::experimental::ranges;

class Non_sortable {};

int main()
{
    auto l = std::vector<Non_sortable>{};
    ranges::sort(ranges::begin(l), ranges::end(l));
}

```

GCC diagnostic

```

<source>: In function 'int main()':
12 : <source>:12:48: error: no matching function for call to 'sort(__gnu_cxx::__norma
    ranges::sort(ranges::begin(l), ranges::end(l));
                                ^
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges

```

```

        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:31:4: note
  I sort(I first, S sent, Comp comp = Comp{}, Proj proj = Proj{})
    ^~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:31:4: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
  from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
  from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/algorithm.hpp:79:15:
  concept bool Sortable =
    ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
  from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
  from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/callable.hpp:187:15:
  concept bool IndirectStrictWeakOrder =
    ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/functio
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/functional.hpp
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
  from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
  from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
  from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:103:15:
  concept bool StrictWeakOrder =
    ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
  concept bool Relation =
    ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:103:15:
  concept bool StrictWeakOrder =
    ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
  concept bool Relation =
    ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:

```

```

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:103:15:
    concept bool StrictWeakOrder =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
    concept bool Relation =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:103:15:
    concept bool StrictWeakOrder =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
    concept bool Relation =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:103:15:
    concept bool StrictWeakOrder =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/function.hpp:75:15:
    concept bool Relation =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:31:4: note
    I sort(I first, S sent, Comp comp = Comp{}, Proj proj = Proj{})
    ^~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:46:23: not
    safe_iterator_t<Rng> sort(Rng&& rng, Comp comp = Comp{}, Proj proj = Proj{})
    ^~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm/sort.hpp:46:23: not
Compiler exited with result code 1

```

Listing 5

Source

```

#include <regex>
#include <type_traits>

```

```

template <typename T>
constexpr bool Destructible = std::is_destructible_v<T>;

template <typename T>
constexpr bool DefaultConstructible = Destructible<T> && std::is_default_constructible_v<T>;

template <typename T>
constexpr bool Movable = Destructible<T> && std::is_move_constructible_v<T> && std::is_copy_constructible_v<T>;

template <typename T>
constexpr bool Copyable = Destructible<T> && std::is_copy_constructible_v<T> && std::is_move_constructible_v<T>;

template <typename T>
constexpr bool Semiregular = Copyable<T> && DefaultConstructible<T>;

template <typename T>
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>(), bool> &&
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;

template <typename T>
constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;

int main()
{
    std::enable_if_t<Regular<std::regex>, std::regex> foo{};
}

```

GCC diagnostic

```

<source>: In instantiation of 'constexpr const bool EqualityComparable<std::__cxx11::
24 : <source>:24:44:   required from 'constexpr const bool Regular<std::__cxx11::basi
28 : <source>:28:20:   required from here
20 : <source>:20:79: error: no match for 'operator==' (operand types are 'std::__cxx1
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:56:0,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:770:5: note: c
    operator==(nullptr_t, const function<_Res(_Args...)>& __f) noexcept
    ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:770:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:56:0,
    from <source>:1:

```

```

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:764:5: note: c
operator==(const function<_Res(_Args...)>& __f, nullptr_t) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:764:5: note:
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:62:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_multimap.h:1045:5: note:
operator==(const multimap<_Key, _Tp, _Compare, _Alloc>& __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_multimap.h:1045:5: note:
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:61:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_map.h:1380:5: note: candi
operator==(const map<_Key, _Tp, _Compare, _Alloc>& __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_map.h:1380:5: note: tem
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:1533:5: note: cand
operator==(const _Rb_tree<_Key, _Val, _KeyOfValue, _Compare, _Alloc>& __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:1533:5: note: te
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:406:5: note: candi
operator==(const _Rb_tree_iterator<_Val>& __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:406:5: note: tem
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre

```

```

        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:908:5: note: candidate: t
operator==(const _Up& __lhs, const optional<_Tp>& __rhs)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:908:5: note:   template a
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:902:5: note: candidate: t
operator==(const optional<_Tp>& __lhs, const _Up& __rhs)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:902:5: note:   template a
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:846:5: note: candidate: t
operator==(nullopt_t, const optional<_Tp>& __rhs) noexcept
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:846:5: note:   template a
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:841:5: note: candidate: t
operator==(const optional<_Tp>& __lhs, nullopt_t) noexcept
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:841:5: note:   template a
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,

```

```

        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:790:5: note: candidate: template<class T>
operator==(const optional<T>& __lhs, const optional<T>& __rhs)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:790:5: note: template argument 1 is deduced from 'char'
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>()>()'
    ^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/vector:64:0,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:51,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_vector.h:1596:5: note: candidate: template<class T, class Alloc>
operator==(const vector<T, Alloc>& __x, const vector<T, Alloc>& __y)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_vector.h:1596:5: note: template argument 1 is deduced from 'char'
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>()>()'
    ^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:61:0,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_stack.h:293:5: note: candidate: template<class T, class Seq>
operator==(const stack<T, Seq>& __x, const stack<T, Seq>& __y)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_stack.h:293:5: note: template argument 1 is deduced from 'char'
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>()>()'
    ^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/deque:64:0,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:60,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:2258:5: note: candidate: template<class T, class Alloc>
operator==(const deque<T, Alloc>& __x, const deque<T, Alloc>& __y)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:2258:5: note: template argument 1 is deduced from 'char'
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>()>()'
    ^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/deque:64:0,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:60,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:274:5: note: candidate: template<class T, class RefL, class PtrL>
operator==(const _Deque_iterator<T, RefL, PtrL>& __x, const _Deque_iterator<T, RefL, PtrL>& __y)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:274:5: note: template argument 1 is deduced from 'char'
20 : <source>:20:79: note: 'std::__cxx11::basic_regex<char>' is not derived from 'constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>()>()'
    ^~~~~~

```

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constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/deque:64:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:267:5: note: cand
operator==(const _Deque_iterator<_Tp, _Ref, _Ptr>& __x,
~~~~~^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:267:5: note:   te
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:378:5: note: can
operator==(nullptr_t, const shared_ptr<_Tp>& __a) noexcept
~~~~~^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:378:5: note:   t
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:373:5: note: can
operator==(const shared_ptr<_Tp>& __a, nullptr_t) noexcept
~~~~~^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:373:5: note:   t
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:368:5: note: can
operator==(const shared_ptr<_Tp>& __a, const shared_ptr<_Up>& __b) noexcept
~~~~~^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:368:5: note:   t
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1420:5: not
operator==(nullptr_t, const __shared_ptr<_Tp, _Lp>& __a) noexcept

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^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1420:5: not
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1415:5: not
operator==(const __shared_ptr<_Tp, _Lp>& __a, nullptr_t) noexcept
^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1415:5: not
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1409:5: not
operator==(const __shared_ptr<_Tp1, _Lp>& __a,
^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1409:5: not
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:694:5: note: can
operator==(nullptr_t, const unique_ptr<_Tp, _Dp>& __x) noexcept
^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:694:5: note:   t
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:689:5: note: can
operator==(const unique_ptr<_Tp, _Dp>& __x, nullptr_t) noexcept
^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:689:5: note:   t
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~

```

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In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:683:5: note: can
    operator==(const unique_ptr<_Tp, _Dp>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:683:5: note:   t
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/tuple:1397:5: note: candidate: tem
    operator==(const tuple<_TElements...>& __t,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/tuple:1397:5: note:   template arg
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/tuple:39:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/array:252:5: note: candidate: temp
    operator==(const array<_Tp, _Nm>& __one, const array<_Tp, _Nm>& __two)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/array:252:5: note:   template argu
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/iterator:66:
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:43,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stream_iterator.h:130:5: note
    operator==(const istream_iterator<_Tp, _CharT, _Traits, _Dist>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stream_iterator.h:130:5: note
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_i
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ios:44,

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        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ostream:38,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/iterator:64,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:43,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/streambuf_iterator.h:204:5: n
    operator==(const istreambuf_iterator<_CharT, _Traits>& __a,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/streambuf_iterator.h:204:5: n
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/ios_bas
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ios:42,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ostream:38,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/iterator:64,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:43,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:311:3: note: candidat
    operator==(const error_condition& __lhs,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:311:3: note:   no kno
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:304:3: note: candidat
    operator==(const error_condition& __lhs, const error_code& __rhs) noexcept
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:304:3: note:   no kno
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:297:3: note: candidat
    operator==(const error_code& __lhs, const error_condition& __rhs) noexcept
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:297:3: note:   no kno
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:292:3: note: candidat
    operator==(const error_code& __lhs, const error_code& __rhs) noexcept
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:292:3: note:   no kno
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5841:5: note:
    operator==(const basic_string<_CharT, _Traits, _Alloc>& __lhs,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5841:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5829:5: note:

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operator==(const _CharT* __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5829:5: note:
20 : <source>:20:79: note:   mismatched types 'const _CharT*' and 'std::__cxx11::basi
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5815:5: note:
operator==(const basic_string<_CharT>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5815:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5807:5: note:
operator==(const basic_string<_CharT, _Traits, _Alloc>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5807:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_s
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:461:5: note: candidate
operator==(__detail::__idnt<basic_string_view<_CharT, _Traits>> __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:461:5: note:   templat
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_s
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:455:5: note: candidate
operator==(basic_string_view<_CharT, _Traits> __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:455:5: note:   templat

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20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_s
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:449:5: note: candidate
    operator==(basic_string_view<_CharT, _Traits> __x,
    ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:449:5: note:   templat
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:41:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:152:5: note: cand
    operator==(const allocator<_Tp>&, const allocator<_Tp>&)
    ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:152:5: note:   te
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:41:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:146:5: note: cand
    operator==(const allocator<_T1>&, const allocator<_T2>&)
    ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:146:5: note:   te
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/char_tr
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:40,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/postypes.h:216:5: note: candi
    operator==(const fpos<_StateT>& __lhs, const fpos<_StateT>& __rhs)
    ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/postypes.h:216:5: note:   tem
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
                                ~~~~~^~~~~~

```

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In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1124:5: note:
    operator==(const move_iterator<_Iterator>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1124:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1118:5: note:
    operator==(const move_iterator<_IteratorL>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1118:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:337:5: note: c
    operator==(const reverse_iterator<_IteratorL>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:337:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:299:5: note: c
    operator==(const reverse_iterator<_Iterator>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:299:5: note:
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/utility:70:0
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:60
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_pair.h:443:5: note: candi

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operator==(const pair<_T1, _T2>& __x, const pair<_T1, _T2>& __y)
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_pair.h:443:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1922:5: note:   candida
operator==(const match_results<_Bi_iter, _Alloc>& __m1,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1922:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1403:5: note:   candida
operator==(const sub_match<_Bi_iter>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1403:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1323:5: note:   candida
operator==(typename iterator_traits<_Bi_iter>::value_type const& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1323:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1249:5: note:   candida
operator==(const sub_match<_Bi_iter>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1249:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1175:5: note:   candida
operator==(typename iterator_traits<_Bi_iter>::value_type const* __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1175:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c

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constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1095:5: note: candidate
      operator==(const sub_match<_Bi_iter>& __lhs,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1095:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1015:5: note: candidate
      operator==(const __sub_match_string<_Bi_iter, _Ch_traits, _Ch_alloc>& __lhs,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1015:5: note:   templ
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:942:5: note: candidat
      operator==(const sub_match<_BiIter>& __lhs, const sub_match<_BiIter>& __rhs)
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:942:5: note:   templa
20 : <source>:20:79: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std
~~~~~^~~~~~
21 : <source>:21:46: error: no match for 'operator!=' (operand types are 'std::__cxx1
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:56:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:788:5: note: c
      operator!=(nullptr_t, const function<_Res(_Args...)>& __f) noexcept
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:788:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:56:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:782:5: note: c
      operator!=(const function<_Res(_Args...)>& __f, nullptr_t) noexcept
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/std_function.h:782:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;

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~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:62:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_multimap.h:1069:5: note:
    operator!=(const multimap<_Key, _Tp, _Compare, _Alloc>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_multimap.h:1069:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:61:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_map.h:1404:5: note: candi
    operator!=(const map<_Key, _Tp, _Compare, _Alloc>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_map.h:1404:5: note:   tem
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:1553:5: note: cand
    operator!=(const _Rb_tree<_Key, _Val, _KeyOfValue, _Compare, _Alloc>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:1553:5: note:   te
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:412:5: note: candi
    operator!=(const _Rb_tree_iterator<_Val>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tree.h:412:5: note:   tem
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:920:5: note: candidate: t
    operator!=(const _Up& __lhs, const optional<_Tp>& __rhs)
    ^~~~~~

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/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:920:5: note: template a
21 : <source>:21:46: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:914:5: note: candidate: t
      operator!=(const optional<_Tp>& __lhs, const _Up& __rhs)
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:914:5: note: template a
21 : <source>:21:46: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:856:5: note: candidate: t
      operator!=(nullopt_t, const optional<_Tp>& __rhs) noexcept
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:856:5: note: template a
21 : <source>:21:46: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:851:5: note: candidate: t
      operator!=(const optional<_Tp>& __lhs, nullopt_t) noexcept
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:851:5: note: template a
21 : <source>:21:46: note: 'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/node_ha
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_tre
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/map:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:52,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:799:5: note: candidate: t
      operator!=(const optional<_Tp>& __lhs, const optional<_Up>& __rhs)
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/optional:799:5: note: template a

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21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/vector:64:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:51,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_vector.h:1620:5: note: ca
      operator!=(const vector<_Tp, _Alloc>& __x, const vector<_Tp, _Alloc>& __y)
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_vector.h:1620:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:61:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_stack.h:317:5: note: cand
      operator!=(const stack<_Tp, _Seq>& __x, const stack<_Tp, _Seq>& __y)
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_stack.h:317:5: note:   te
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/deque:64:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:2284:5: note: can
      operator!=(const deque<_Tp, _Alloc>& __x,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:2284:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/deque:64:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:287:5: note: cand
      operator!=(const _Deque_iterator<_Tp, _RefL, _PtrL>& __x,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:287:5: note:   te
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/deque:64:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/stack:60,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:47,
      from <source>:1:

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/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:280:5: note: cand
operator!=(const _Deque_iterator<_Tp, _Ref, _Ptr>& __x,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_deque.h:280:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:393:5: note: can
operator!=(nullptr_t, const shared_ptr<_Tp>& __a) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:393:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:388:5: note: can
operator!=(const shared_ptr<_Tp>& __a, nullptr_t) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:388:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81:0,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:383:5: note: can
operator!=(const shared_ptr<_Tp>& __a, const shared_ptr<_Up>& __b) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr.h:383:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81,
from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1436:5: not
operator!=(nullptr_t, const __shared_ptr<_Tp, _Lp>& __a) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1436:5: not
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_

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        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1431:5: not
operator!=(const __shared_ptr<_Tp, _Lp>& __a, nullptr_t) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1431:5: not
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/memory:81,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:45,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1425:5: not
operator!=(const __shared_ptr<_Tp1, _Lp>& __a,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/shared_ptr_base.h:1425:5: not
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:711:5: note: can
operator!=(nullptr_t, const unique_ptr<_Tp, _Dp>& __x) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:711:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:706:5: note: can
operator!=(const unique_ptr<_Tp, _Dp>& __x, nullptr_t) noexcept
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:706:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:700:5: note: can
operator!=(const unique_ptr<_Tp, _Dp>& __x,

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^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_ptr.h:700:5: note:   t
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/tuple:1423:5: note: candidate: tem
      operator!=(const tuple<_TElements...>& __t,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/tuple:1423:5: note:   template arg
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/tuple:39:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/unique_
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/locale:43,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:44,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/array:257:5: note: candidate: temp
      operator!=(const array<_Tp, _Nm>& __one, const array<_Tp, _Nm>& __two)
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/array:257:5: note:   template argu
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/iterator:66:
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:43,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stream_iterator.h:137:5: note
      operator!=(const istream_iterator<_Tp, _CharT, _Traits, _Dist>& __x,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stream_iterator.h:137:5: note
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/locale_
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_i
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ios:44,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ostream:38,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/iterator:64,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:43,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/streambuf_iterator.h:210:5: n
      operator!=(const istreambuf_iterator<_CharT, _Traits>& __a,

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^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/streambuf_iterator.h:210:5: n
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/ios_bas
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ios:42,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/ostream:38,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/iterator:64,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:43,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:331:3: note: candidat
      operator!=(const error_condition& __lhs,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:331:3: note:   no kno
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:327:3: note: candidat
      operator!=(const error_condition& __lhs, const error_code& __rhs) noexcept
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:327:3: note:   no kno
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:323:3: note: candidat
      operator!=(const error_code& __lhs, const error_condition& __rhs) noexcept
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:323:3: note:   no kno
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:319:3: note: candidat
      operator!=(const error_code& __lhs, const error_code& __rhs) noexcept
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/system_error:319:3: note:   no kno
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5879:5: note:
      operator!=(const basic_string<_CharT, _Traits, _Alloc>& __lhs,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5879:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5867:5: note:
      operator!=(const _CharT* __lhs,
      ^~~~~~

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5867:5: note:
21 : <source>:21:46: note:   mismatched types 'const _CharT*' and 'std::__cxx11::basi
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~

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In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5854:5: note:
operator!=(const basic_string<_CharT, _Traits, _Alloc>& __lhs,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_string.h:5854:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_s
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:479:5: note: candidate
operator!=(__detail::__idnt<basic_string_view<_CharT, _Traits>> __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:479:5: note:   templat
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_s
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:473:5: note: candidate
operator!=(basic_string_view<_CharT, _Traits> __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:473:5: note:   templat
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/basic_s
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:52,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:467:5: note: candidate
operator!=(basic_string_view<_CharT, _Traits> __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string_view:467:5: note:   templat
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
    ~~~~~~^~~~~~

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:41:0,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,

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        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:164:5: note: cand
operator!=(const allocator<_Tp>&, const allocator<_Tp>&)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:164:5: note:   te
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
        ~~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:41:0,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:158:5: note: cand
operator!=(const allocator<_T1>&, const allocator<_T2>&)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/allocator.h:158:5: note:   te
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
        ~~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/char_tr
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/string:40,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bitset:47,
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:39,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/postypes.h:221:5: note: candi
operator!=(const fpos<_StateT>& __lhs, const fpos<_StateT>& __rhs)
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/postypes.h:221:5: note:   tem
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
        ~~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1136:5: note:
operator!=(const move_iterator<_Iterator>& __x,
    ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1136:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
        ~~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
        from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
        from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1130:5: note:
operator!=(const move_iterator<_IteratorL>& __x,

```

```

^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:1130:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:349:5: note: c
      operator!=(const reverse_iterator<_IteratorL>& __x,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:349:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_alg
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:61
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:311:5: note: c
      operator!=(const reverse_iterator<_Iterator>& __x,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_iterator.h:311:5: note:
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/utility:70:0
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:60
      from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_pair.h:456:5: note: candi
      operator!=(const pair<_T1, _T2>& __x, const pair<_T1, _T2>& __y)
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_pair.h:456:5: note: tem
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1946:5: note: candida
      operator!=(const match_results<_Bi_iter, _Alloc>& __m1,
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1946:5: note: templ
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
      from <source>:1:

```

```

/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1419:5: note: candidate
operator!=(const sub_match<_Bi_iter>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1419:5: note:   templ
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1339:5: note: candidate
operator!=(typename iterator_traits<_Bi_iter>::value_type const& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1339:5: note:   templ
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1262:5: note: candidate
operator!=(const sub_match<_Bi_iter>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1262:5: note:   templ
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1188:5: note: candidate
operator!=(typename iterator_traits<_Bi_iter>::value_type const* __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1188:5: note:   templ
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1111:5: note: candidate
operator!=(const sub_match<_Bi_iter>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1111:5: note:   templ
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1031:5: note: candidate
operator!=(const __sub_match_string<_Bi_iter, _Ch_traits, _Ch_alloc>& __lhs,
^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:1031:5: note:   templ

```

```

21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 's
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:62:0,
      from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:953:5: note: candidat
      operator!=(const sub_match<_BiIter>& __lhs, const sub_match<_BiIter>& __rhs)
      ^~~~~~
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/regex.h:953:5: note:   templa
21 : <source>:21:46: note:   'std::__cxx11::basic_regex<char>' is not derived from 'c
      std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;
      ~~~~~^~~~~~
Compiler exited with result code 1

```

Clang diagnostic

```

20 : <source>:20:79: error: invalid operands to binary expression ('std::__1::basic_r
      constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std:
      ~~~~~^~~~~
24 : <source>:24:44: note: in instantiation of variable template specialization 'Equa
      constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;
      ^
28 : <source>:28:20: note: in instantiation of variable template specialization 'Regu
      std::enable_if_t<Regular<std::regex>, std::regex> foo{};
      ^
24 : <source>:24:44: error: constexpr variable 'Regular<std::__1::basic_regex<char, s
      constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;
      ~~~~~^~~~~~
28 : <source>:28:20: note: in instantiation of variable template specialization 'Regu
      std::enable_if_t<Regular<std::regex>, std::regex> foo{};
      ^
28 : <source>:28:20: error: non-type template argument is not a constant expression
      std::enable_if_t<Regular<std::regex>, std::regex> foo{};
      ^~~~~~
28 : <source>:28:20: note: initializer of 'Regular<std::__1::basic_regex<char, std::__
24 : <source>:24:16: note: declared here
      constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;
      ^
3 errors generated.
Compiler exited with result code 1

```

MSVC diagnostic

example.cpp

```
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/xlocale(314): warning C
20 : <source>(20): error C2678: binary '==': no operator found which takes a left-han
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/exception(336): note: c
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/exception(341): note: o
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/exception(346): note: o
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(362): note
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(370): note
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(378): note
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(386): note
20 : <source>(20): note: while trying to match the argument list '(std::regex, std::r
24 : <source>(24): note: see reference to variable template 'const bool EqualityCompa
28 : <source>(28): note: see reference to variable template 'const bool Regular<std::
20 : <source>(20): error C2065: 'is_same_v': undeclared identifier
21 : <source>(21): error C2678: binary '!=': no operator found which takes a left-han
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/exception(351): note: c
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/exception(356): note: o
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/exception(361): note: o
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(395): note
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(402): note
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(409): note
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/system_error(416): note
21 : <source>(21): note: while trying to match the argument list '(std::regex, std::r
21 : <source>(21): error C2065: 'is_same_v': undeclared identifier
20 : <source>(20): error C2131: expression did not evaluate to a constant
20 : <source>(20): note: failure was caused by non-constant arguments or reference to
20 : <source>(20): note: see usage of 'is_same_v'
28 : <source>(28): error C2970: 'std::enable_if_t': template parameter '_Test': 'Regu
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/type_traits(2099): note
24 : <source>(24): note: see declaration of 'Regular<std::regex>'
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/type_traits(2099): erro
    with
    [
        _Ty=std::regex
    ]
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/type_traits(2099): note
    with
    [
        _Ty=std::regex
    ]
28 : <source>(28): note: see reference to alias template instantiation 'enable_if_t<f
/opt/compiler-explorer/windows/19.10.25017/lib/native/include/type_traits(2099): erro
28 : <source>(28): error C2938: 'enable_if_t<false,std::regex>' : Failed to specializ
Microsoft (R) C/C++ Optimizing Compiler Version 19.10.25017 for x64
Copyright (C) Microsoft Corporation. All rights reserved.
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
```

```

err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
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err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
Compiler exited with result code 2

```

Listing 6

Source

```

#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <regex>

namespace ranges = std::experimental::ranges;
using ranges::Regular;

int main()
{
    Regular foo = std::regex{};
}

```

GCC diagnostic

```

<source>: In function 'int main()':
10 : <source>:10:28: error: deduced initializer does not satisfy placeholder constraint
    Regular foo = std::regex{};
                        ^
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/regular.hpp:2
    concept bool Regular =
        ^~~~~~

```

```

In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:94:15: n
    concept bool EqualityComparable =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
    concept bool WeaklyEqualityComparable =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/compare.hpp:74:15: n
Compiler exited with result code 1

```

Listing 7

Source

```

#include <regex>
#include <type_traits>

class Foo {
public:
    Foo() = delete;
};

constexpr bool operator==(const Foo, const Foo) noexcept;
constexpr bool operator!=(const Foo, const Foo) noexcept;

template <typename T>
constexpr bool Destructible = std::is_destructible_v<T>;

template <typename T>
constexpr bool DefaultConstructible = Destructible<T> && std::is_default_constructible_v<T>;

template <typename T>
constexpr bool Movable = Destructible<T> && std::is_move_constructible_v<T> && std::is_move_assignable_v<T>;

template <typename T>

```

```
constexpr bool Copyable = Destructible<T> && std::is_copy_constructible_v<T> && std::

template <typename T>
constexpr bool Semiregular = Copyable<T> && DefaultConstructible<T>;

template <typename T>
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>()), bool>;

template <typename T>
constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;

int main()
{
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
}
```

GCC diagnostic

```
In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/move.h:
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_pai
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/utility:70,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:60
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/type_traits: In substitution of 't
36 : <source>:36:37:   required from here
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/type_traits:2479:61: error: no typ
    using enable_if_t = typename enable_if<_Cond, _Tp>::type;
                                   ^
<source>: In function 'int main()':
36 : <source>:36:49: error: cannot convert 'Foo' to 'int' in initialization
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
                                   ^
Compiler exited with result code 1
```

Clang diagnostic

```
In file included from <source>:1:
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/regex:
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/stdexc
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/except
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/cstdde
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/type_traits:427:78: error: n
```

```
template <bool _Bp, class _Tp = void> using enable_if_t = typename enable_if<_Bp, _Tp>::type;
36 : <source>:36:8: note: in instantiation of template type alias 'enable_if_t' requested from <source>:36:8
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
    ^
1 error generated.
Compiler exited with result code 1
```

MSVC diagnostic

[illegible]

Listing 8

Source

```
#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <regex>

namespace ranges = std::experimental::ranges;
using ranges::Regular;

class Foo {
public:
    Foo() = delete;
};

constexpr bool operator==(const Foo, const Foo) noexcept;
constexpr bool operator!=(const Foo, const Foo) noexcept;

int main()
{
    Regular foo = Foo{};
}
```

GCC diagnostic

```
<source>: In function 'int main()':
18 : <source>:18:21: error: deduced initializer does not satisfy placeholder constraint
    Regular foo = Foo{};
                  ^

In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/regular.hpp:2
    concept bool Regular =
        ^~~~~~

In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorithm
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/semiregular.h
```

```

    concept bool Semiregular =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool DefaultConstructible =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
<source>: At global scope:
13 : <source>:13:16: warning: inline function 'constexpr bool operator==(Foo, Foo)' u
    constexpr bool operator==(const Foo, const Foo) noexcept;
        ^~~~~~
14 : <source>:14:16: warning: inline function 'constexpr bool operator!=(Foo, Foo)' u
    constexpr bool operator!=(const Foo, const Foo) noexcept;
        ^~~~~~
Compiler exited with result code 1

```

Listing 9

Source

```

#include <regex>
#include <type_traits>

class Foo {
public:
    Foo() = delete;
    Foo(const Foo&) = delete;
};

```

```

constexpr bool operator==(const Foo&, const Foo&) noexcept;
constexpr bool operator!=(const Foo&, const Foo&) noexcept;

template <typename T>
constexpr bool Destructible = std::is_destructible_v<T>;

template <typename T>
constexpr bool DefaultConstructible = Destructible<T> && std::is_default_constructible_v<T>;

template <typename T>
constexpr bool Movable = Destructible<T> && std::is_move_constructible_v<T> && std::is_move_assignable_v<T>;

template <typename T>
constexpr bool Copyable = Destructible<T> && std::is_copy_constructible_v<T> && std::is_copy_assignable_v<T>;

template <typename T>
constexpr bool Semiregular = Copyable<T> && DefaultConstructible<T>;

template <typename T>
constexpr bool EqualityComparable = std::is_same_v<decltype(std::declval<T>()) == std::declval<T>(), bool> &&
    std::is_same_v<decltype(std::declval<T>()) != std::declval<T>(), bool>;

template <typename T>
constexpr bool Regular = Semiregular<T> && EqualityComparable<T>;

int main()
{
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
}

```

GCC diagnostic

```

In file included from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/move.h:
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/bits/stl_pair.h:77,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/utility:70,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/algorithm:60,
    from /opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/regex:38,
    from <source>:1:
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/type_traits: In substitution of 't
37 : <source>:37:37:   required from here
/opt/compiler-explorer/gcc-7.2.0/include/c++/7.2.0/type_traits:2479:61: error: no typ
    using enable_if_t = typename enable_if<_Cond, _Tp>::type;
                                                                    ^
<source>: In function 'int main()':
37 : <source>:37:49: error: cannot convert 'Foo' to 'int' in initialization
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};

```

$\wedge$

Clang diagnostic

```
In file included from <source>:1:
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/regex:
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/stdexch:
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/except:
In file included from /opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/cstdde:
/opt/compiler-explorer/clang-5.0.0/bin/./include/c++/v1/type_traits:427:78: error: n
template <bool _Bp, class _Tp = void> using enable_if_t = typename enable_if<_Bp, _Tp
                                                                    ^~~
37 : <source>:37:8: note: in instantiation of template type alias 'enable_if_t' requ
    std::enable_if_t<Regular<Foo>, Foo> foo = Foo{};
        ^
1 error generated.
Compiler exited with result code 1
```

MSVC diagnostic

[illegible]

```
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
err:msvcrt:demangle_datatype Unknown type s
Compiler exited with result code 2
```

Listing 10

Source

```
#include <experimental/ranges/algorithm>
#include <experimental/ranges/iterator>
#include <regex>

namespace ranges = std::experimental::ranges;
using ranges::Regular;

class Foo {
public:
    Foo() = delete;
    Foo(const Foo&) = delete;
};

constexpr bool operator==(const Foo&, const Foo&) noexcept;
constexpr bool operator!=(const Foo&, const Foo&) noexcept;

int main()
{
    Regular foo = Foo{};
}
```

GCC diagnostic

```
<source>: In function 'int main()':
19 : <source>:19:21: error: deduced initializer does not satisfy placeholder constraint
    Regular foo = Foo{};
                  ^
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
```

```

        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/regular.hpp:2
    concept bool Regular =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/semiregular.h
    concept bool Semiregular =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/semiregular.h
    concept bool Copyable =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/semiregular.h
    concept bool CopyConstructible =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool MoveConstructible =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
    concept bool ConvertibleTo =
        ^~~~~~

```

```

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~

In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~

In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~

/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges

```

```

        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
    concept bool ConvertibleTo =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
    concept bool ConvertibleTo =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
    concept bool ConvertibleTo =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
    concept bool ConvertibleTo =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concept
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit

```

```

        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/movable.hpp:2
    concept bool Movable =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool MoveConstructible =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:98:15: note
    concept bool ConvertibleTo =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hpp:108:15: note: within
    concept bool Swappable =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hpp:108:15: note:      wi
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hpp:108:15: note:      wi
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hpp:108:15: note: the re
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.

```

```

        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
        from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
        from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
        from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool DefaultConstructible =
        ^~~~~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/object/move_construc
    concept bool Constructible =
        ^~~~~~
In file included from /opt/compiler-explorer/libs/cmcstl2/include/stl2/type_traits.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/swap.hp
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/tagged.
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/algorit
    from /opt/compiler-explorer/libs/cmcstl2/include/stl2/algorithm.hpp:
    from /opt/compiler-explorer/libs/cmcstl2/include/experimental/ranges
    from <source>:1:
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
    concept bool _Is = _Valid<T, U, V...> && __bool<T<U, V...>::value>;
        ^~~
/opt/compiler-explorer/libs/cmcstl2/include/stl2/detail/concepts/core.hpp:30:15: note
compilation terminated due to -fmax-errors=1.
Compiler exited with result code 1

```